Assignment 4 | 1905074

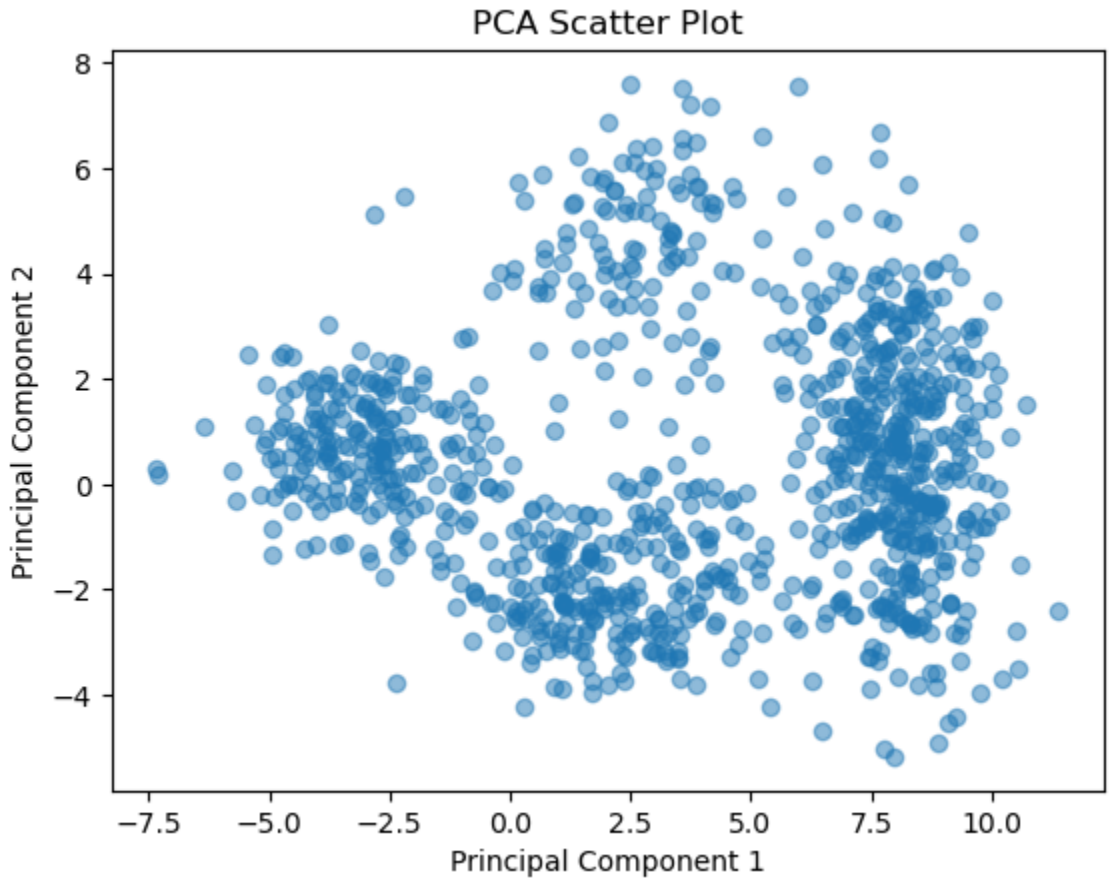
How to run Code:

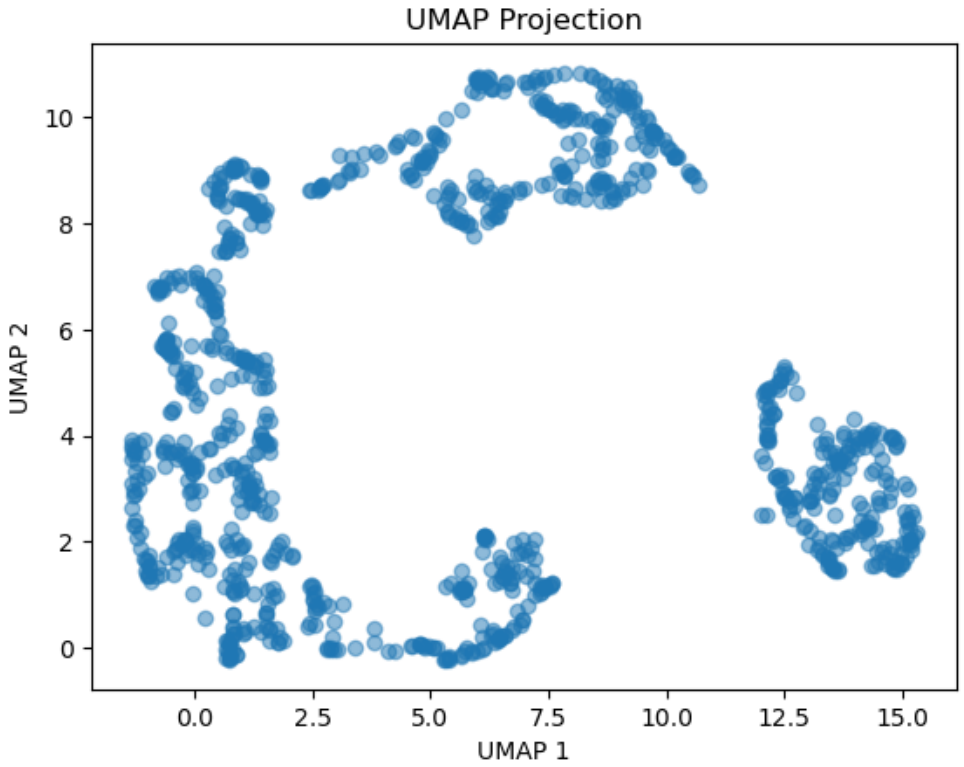
PCA: python -u "path to parent folder \1905074\codes\1905074_PCA.py"

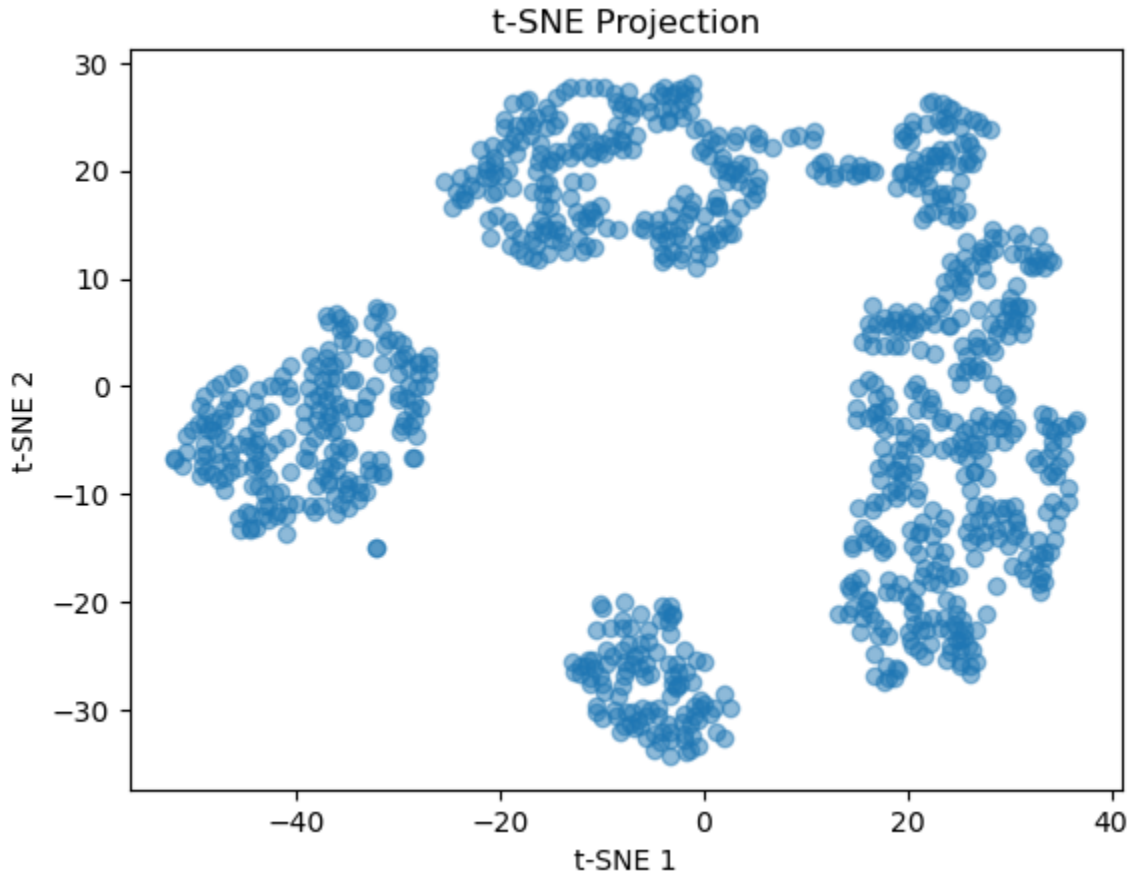
EM: python -u " path to parent folder \1905074\1905074_EM.py"

Note: keep the input files in the codes folder

PCA Plots:




UMAP Projection
UMAP 2
UMAP 1
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0


t-SNE Projection
t-SNE 2
t-SNE 1
30
20
10
0
−10
−20
−30
−40
−20
0
20
40

EM Algorithm result:

Mean number of children in families with family planning: 1.7889842647985699

Mean number of children in families without family planning: 4.916301802525844

Proportion of families with family planning: 0.35791114558174286

Proportion of families without family planning: 0.6420888544182571